Uncertainty Evaluation in Engineering Measurements and Machine Learning

Instructor: *Miodrag Bolic, University of Ottawa, Ottawa, Canada*

October 5, 2026

Course Information

Time and place: Fall 2026, Monday 2:30 pm – 5:30 pm, 550 Cumberland (TBT) 323
Course code: ELG 5218, CSI 5218, EACJ 5600
Github page: https://github.com/Health-Devices/Course-Uncertainty-Machine-Learning-2026
Contact: Miodrag Bolic's email, Xuanyu Su's email

Calendar Style Description: Uncertainty, uncertainty propagation, Bayesian inference, Bayesian filtering, data fusion, metrology, measurement science, error analysis, measures of agreement, data quality, and data quality index. Case studies will be drawn from various fields, including biomedical instrumentation, sensors and signal processing.

Prerequisites

We expect participating students to bring basic knowledge and experience in:

- Elementary probability
- Elementary statistics
- Machine learning

Grading

For collecting the credits the students are expected to complete:

- Assignments or projects (30% of the grade)
- Midterms (30% of the grade), on October 19 and November 16.
- Final exam (40% of the grade)

Course organizing principle

This course begins with *measurement uncertainty* and progressively builds modern uncertainty-aware ML:

- **Bayesian modeling** (modeling simple problems),

- **Classical probabilistic inference** (IID and sequential),
- **Latent-variable models** (state-space models, Kalman filters)
- **Uncertainty scores** (proper scores, calibration, conformal prediction),
- **Modern architectures** (multimodal fusion, diffusion models, SciML),
- **Sequential decision making** (intro to reinforcement learning).

Course materials will be provided as weekly slides and Jupyter notebooks. Each week, the notebooks will be posted in advance of (or shortly after) the lecture and will include the key concepts, derivations, examples, and implementation exercises covered that week. Students are expected to review the notebooks regularly, work through the examples, and ensure they understand the underlying concepts. The notebooks are an essential component of the course and will be used to support assignments, discussions, and exam preparation.

Lecture plan

The 12 lectures are shown in Table 1. Please note that the lecture plan is preliminary and will likely change during the semester.

Table 1: Preliminary lecture plan.

L#	Theme	Detailed topics
L1	Bayesian modeling and likelihoods	Measurement models; MLE/MAP; identify aleatoric vs epistemic uncertainty
L2	Bayesian regression and classification	Compute predictive distributions; intervals; detection/classification
L3	Monte Carlo, bootstrap, propagation	Propagate uncertainty through pipelines; bootstrap uncertainty; sensitivity basics
L4	State-space models	Kalman/HMM fundamentals; interpret covariance evolution; Particle filters;
L5	Deep learning + latent representations	VAE/ELBO; connect state-space and deep learning
L6	Uncertainty in deep nets	Ensembles, MC dropout, Laplace, approx. BNNs
L7	Proper scores and calibration	NLL/Brier/CRPS; reliability diagrams; ECE; temperature scaling
L8	Conformal prediction	Build prediction intervals/sets with finite-sample coverage; efficiency analysis
L9	Multimodal fusion with uncertainty	Encoders + fusion; UQ wrappers
L10	Diffusion models I (DDPM)	Implement forward noising + denoising loss; reverse sampling intuition
L11	Scientific ML + out-of-distribution (OOD)	PINNs/operators/inverse problems; OOD detection and drift monitoring
L12	Sequential decision making	Uncertainty in sequential decision making, POMDP models

Why Uncertainty Quantification?

Over the last several years, deep neural networks have advanced many applications, including vision, language understanding, speech understanding, robotics, and more. But a major challenge still remains: how to model uncertainty. Good models of uncertainty are crucial whenever decisions need to be made or an algorithm must decide when and how to acquire new information. Uncertainty quantification involves combining computational models, physical observations, and, if applicable, expert judgment to infer about a physical system. Types of uncertainties include aleatoric and epistemic uncertainties as shown in Figure 1.

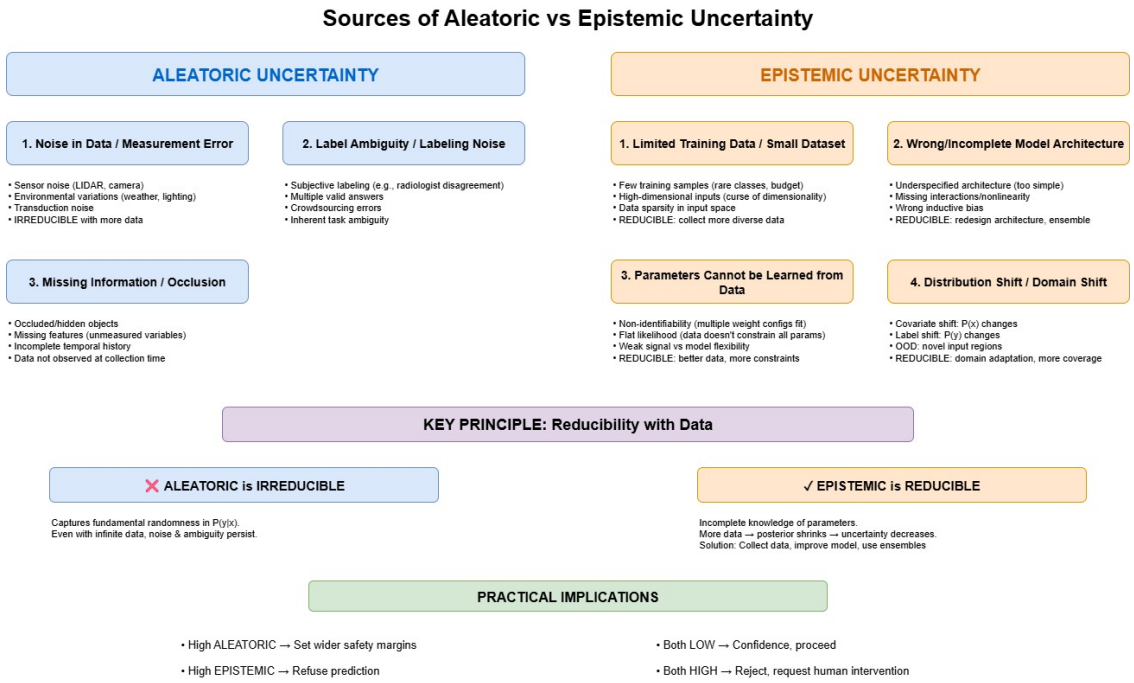


Figure 1: Sources of Aleatoric vs Epistemic Uncertainty

Use of Generative AI Tools

(This section is modified from UToronto CSC311 - Introduction to Machine Learning (Fall 2024))

You may use generative AI tools (e.g., ChatGPT, Copilot) as learning aids, but they are not required for this course. Treat them as supplementary resources for exploration—not as substitutes for your own work. You are fully responsible for your learning and for all submitted work.

Important Rules

- No AI use on exams (midterm or final).
- Assignments are designed to be completed without AI tools using course concepts.
- Overreliance on AI can harm your understanding and lead to poor performance later.

Risks and Warnings

- AI tools may produce incorrect or misleading content, false citations, or violate privacy and security standards.
- Submitting false citations or uncredited AI-generated content is an academic offense.

Permitted Uses

- Answer general conceptual questions.
- Provide examples of library/API usage.
- Summarize information.
- Generate test cases.
- Assist with debugging.

Prohibited Uses

- Copying AI-generated text or code into your submission without acknowledgment.
- Using AI beyond the permitted purposes.

Documentation Requirements

- Include a statement in your submission describing which tool you used and how.

Links

Relevant Courses

- Our course in 2021
- Bayesian Machine Learning and Information Processing (5SSD0) by Prof. dr. ir. Bert de Vries
- Advanced Bayesian Learning by Mattias Villani
- Probabilistic Machine Learning (Summer 2020)
- Prince, Simon J.D. (2025). Understanding Deep Learning

Books

- Murphy, Kevin P. (2021). *Probabilistic Machine Learning: An Introduction*. MIT Press.
- Murphy, Kevin P. (2022). *Probabilistic Machine Learning: Advanced Topics*.
- Izbicki, Rafael (2025). *Machine Learning Beyond Point Predictions: Uncertainty Quantification,* `https://github.com/rizbicki/UQ4ML`.
- Villani, Mattias, (2025). *Bayesian Learning.*
- Prince, Simon J.D. (2025). Understanding Deep Learning

- Theodoridis, Sergios (2025) Machine Learning: From the Classics to Deep Networks, Transformers, and Diffusion Models
- Barber, David. *Bayesian Reasoning and Machine Learning*.
- Davidson-Pilon, Cameron. *Bayesian Methods for Hackers*.
- Goodman, N. D., Tenenbaum, J. B., and The ProbMods Contributors (2016). *Probabilistic Models of Cognition* (2nd ed.).
- Winn, John. *Model-Based Machine Learning*: https://www.mbmlbook.com/
- Cinelli, L. P. et al. (2021). *Variational Methods for Machine Learning with Applications to Deep Networks*.
- Thuerey, N. et al. (Dec 2021). *Physics-based Deep Learning*: http://physicsbaseddeeplearning.org
- Martin, Osvaldo et al. (2021). *Bayesian Modeling and Computation in Python*.
- Yadav, Neha; Yadav, Anupam (2015). *An Introduction to Neural Network Methods for Differential Equations*.
- Goulet, James-A. (2020). *Probabilistic Machine Learning for Civil Engineers*: http://profs.polymtl.ca/jagoulet/Site/Goulet_web_page_BOOK.html
- ME 539 - Introduction to Scientific Machine Learning: https://github.com/PredictiveScienceLab/data-analytics-se?_ga=2.66952878.1678226653.1765915278-1039759646.1765915278

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